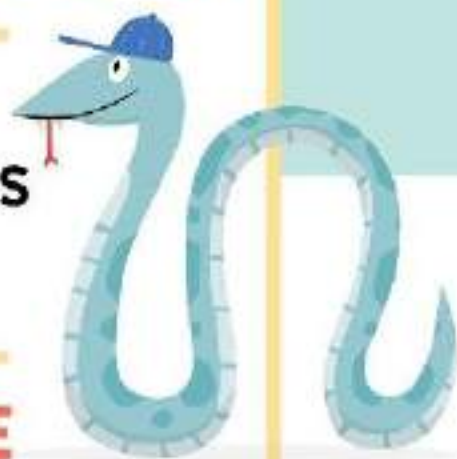


CODING FOR KIDS PYTHON

LEARN TO CODE WITH
50 AWESOME GAMES
AND ACTIVITIES

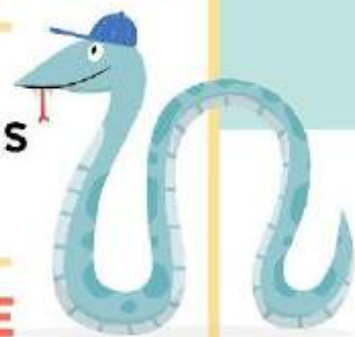
ADRIENNE B. TACKE



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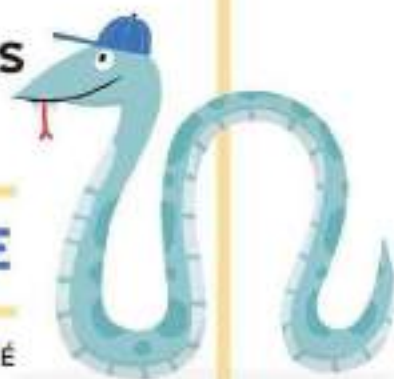


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ILLUSTRATIONS BY AMIR ABOU ROUMIÉ




ROCKRIDGE
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To the technologists of tomorrow



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INTRODUCTION

***Coding for Kids: Python* is a unique and fun introduction** to the Python programming language. Written for someone with absolutely no experience with coding, this book uses silly analogies, helpful examples, and many activities and games to help anyone learn how to code in Python!

Let me share a little about myself, your excited author: I am currently a full-time software engineer who finds joy and fulfillment in helping new and potential coders of all kinds. I've spent many hours volunteering at local elementary and high schools, speaking to students about careers in software development, and teaching the basics of coding. It is so fun to see the spark in someone's eye when they see the power of code. I wrote this book to spark the imagination and wonder of many more people!

Code is at the core of almost everything we use and love. We can write code to make games, create music and art, bring robots to life, and power almost anything that is electronic. When you realize how much coding is a part of the world of the future, it becomes so important to learn how it works! This book will help you do just that.

Coding is literally translating human ideas and actions into a language that computers can understand. Python is one of the languages, but there are many others—for example: JavaScript, C#, Ruby, and C++. Each of these languages tells the computer how to do something, but each one does it a bit differently. I chose Python for this book because it is very close to our speaking language of English, which makes the coding concepts I will introduce a little easier to understand. :)

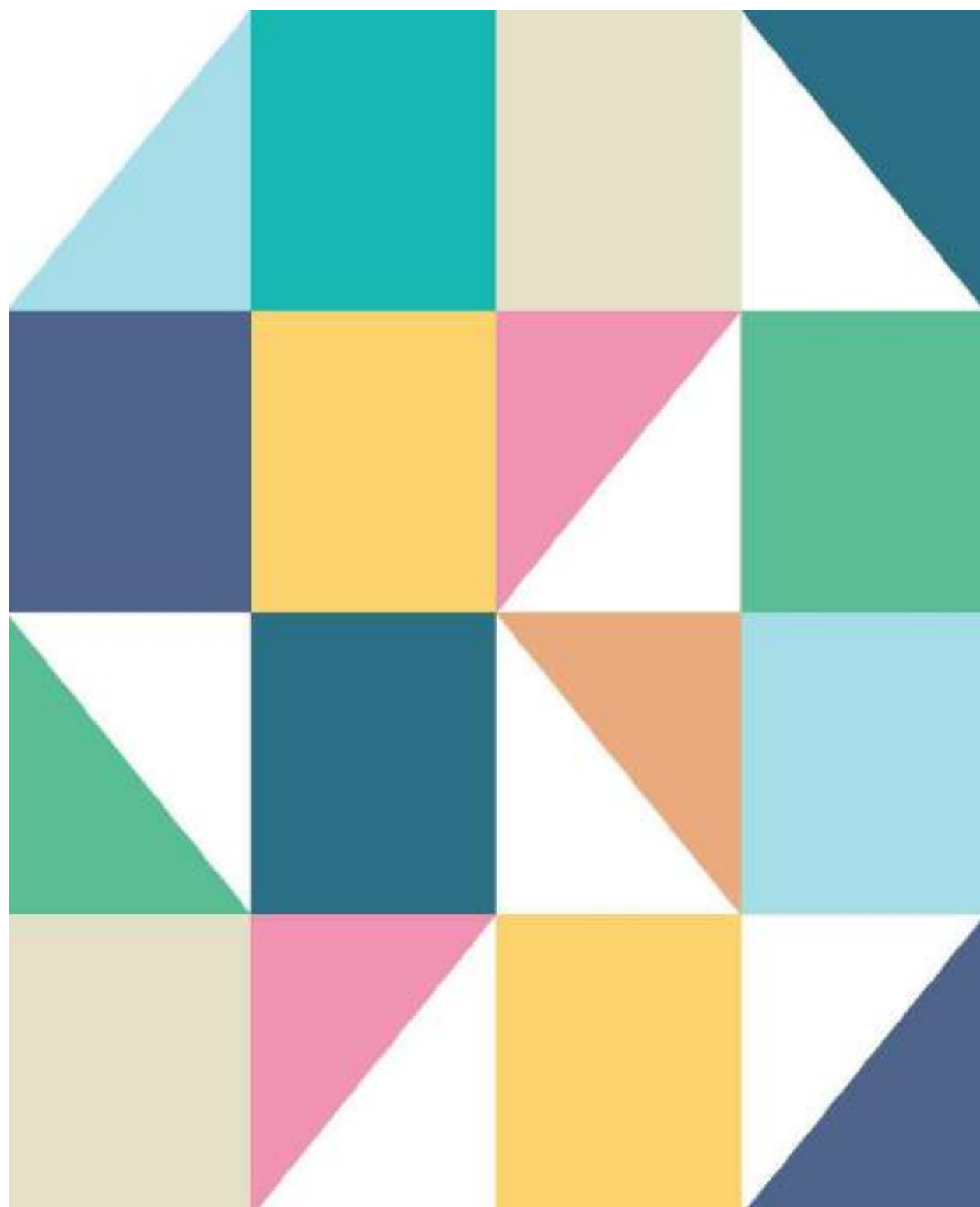
The best part about coding is that you just need a computer and this book! From the first chapter to the last, I'll guide you through the coding concepts you need to know, with step-by-step instructions and examples, plenty of helpful screenshots, and simple

explanations of the new programming terms you'll learn (all of which are also available for you in the handy [glossary](#) at the back of the book!). By the time you've finished this book, you'll be creating some really cool programs and even simple games that you can play with friends! Hangman, anyone?

Coding is one skill that you'll only learn by doing. That's why I've structured each chapter to walk you through the code as you follow along. This makes the book truly interactive, as you'll learn about a concept, write some code, understand what it's doing, read and learn a bit more, maybe fix a bug or two, and see the results of your code in real time! And to really help you understand the different coding concepts in this book, I've also included activities at the end of each chapter to help you test your knowledge, combine multiple concepts, and write more code. After all, practice makes perfect—especially with coding! Finally, if some of the activities are too easy for you, or if you just want to keep coding, I've included even more difficult challenges after each chapter to really stretch your brain and give you more chances to show off your creativity!

This book will help you start an incredible adventure into the world of programming. By the end of it, you'll be ready for the world of the future! What are you waiting for?







WELCOME TO PYTHON!

Hey there! Since you're reading this book, you must be a pretty curious and cool person. Why? Because you want to learn how to code! And who wouldn't? Coding is an awesome skill that can help you build all kinds of things and solve a lot of problems. When you code, you take human ideas and then translate them into a language that a machine can understand.

Coding is built around the concept of input and output. We give the computer some **input**, which is any information or data provided by us humans, and expect some **output**, which can be words, pictures, an action, or some other result, after the computer has processed the input we gave it. Sounds interesting, doesn't it?

How many things can you think of that follow this input/output, or **I/O**, concept? For example, when we press buttons on a controller or swipe our fingers left and right on a mobile game, that's input. And when our character jumps, ducks, moves left, or moves right, that's output. How about baking? All of the ingredients we need to make cookies can be considered input. After following the instructions and using the ingredients, we get our output, which are the baked cookies!

Using examples like these, as well as silly scenarios and conversations with a computer, we'll explore how to code in Python, and you'll understand what we're doing in no time! The coolest thing about coding is that you can do it from almost anywhere. All you

need is a laptop or computer (Windows PC or Mac is fine), and I'll help you with the rest.

Are you ready to learn how to speak to a computer? Hooray! Let's go!



WHY PYTHON?

Just like humans can understand many different languages, a computer can understand the ideas and concepts that we input to it through several different programming languages. In this book, we'll focus on the Python programming language because Python is easy to understand, can be used in many different ways, and is quick to learn. Also, it is a popular language that runs on almost every machine and is used at many big, important organizations like Google, Instagram, NASA, and Spotify.

INSTALLING PYTHON

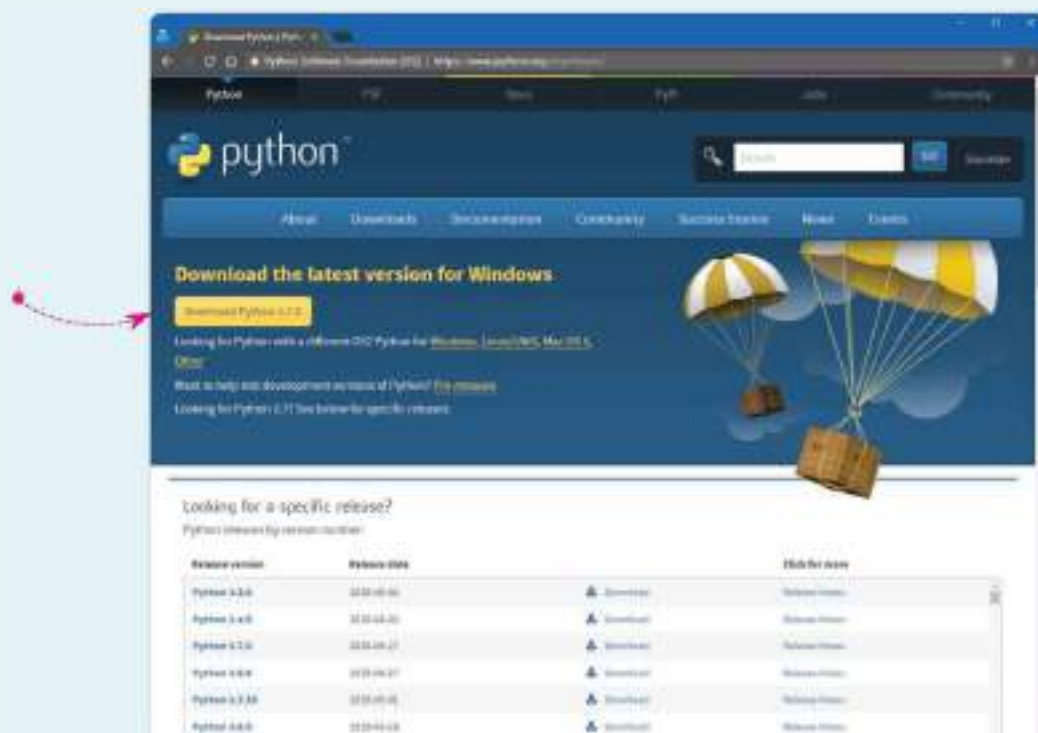
I know we want to dive right into coding, but we can't do that until we have the right tools. I'll walk you through the step-by-step process of installing Python. Let's get started!

ON A PC

If you are on a Windows machine, you probably don't have Python installed already. This is because Windows operating systems don't

usually come with the Python language. That's okay, though! We can get it ourselves. :)

1. On your computer, open an Internet browser like Google Chrome or Mozilla Firefox.
2. In the address bar, type "<https://www.python.org/downloads/>" to go to the official Python website's Downloads section.
3. Through the magic of coding, the website will probably know what type of computer you are using, and the **DOWNLOAD** button will show you the correct version of Python to install! In our case, we want the latest version, which was Python 3.7.0 when I wrote the book. Don't worry if it tells you to download a newer version. Go ahead and click the **DOWNLOAD** button.



4. A download will start and will probably go to the bottom of your window like in the picture.



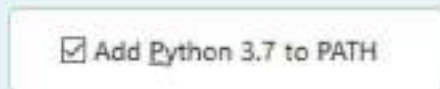
- 5 . Once your download is complete, click on it to begin the installation. When you do, a window should pop up.



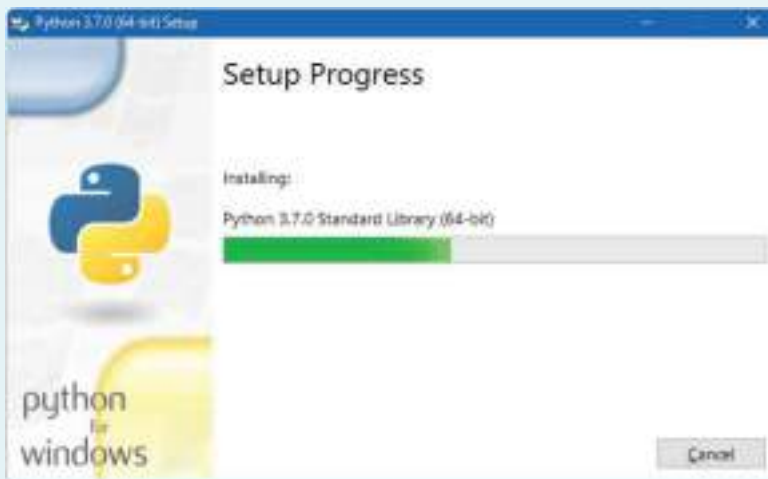
6. Go ahead and click the **RUN** button. Then, you'll see this window (yours may say 32-bit if that's right for your machine):



- 7 . Make sure to check the **ADD PYTHON 3.7 TO PATH** checkbox.



8. Click **INSTALL NOW** . Python should begin installing. You should see a window like this one:



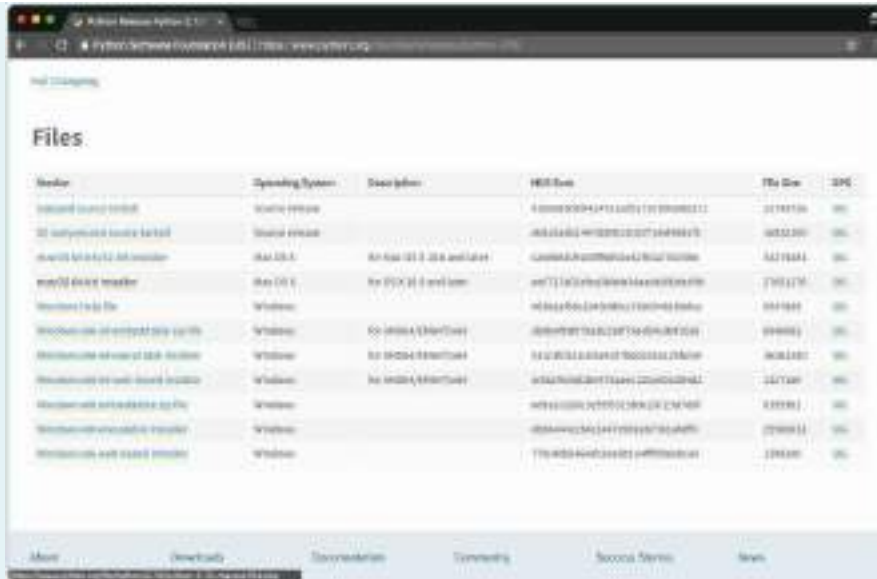
9. Wait for the install to finish and the green bar to be full. Once it is done, the final screen should appear, saying that your installation was successful.



10. Whoo-hoo! You're done! Click the **CLOSE** button and you're ready to go. You've installed Python on Windows !

ON A MAC

1. On your computer, open an Internet browser like Google Chrome or Mozilla Firefox.
2. In the address bar, type "<https://www.python.org/downloads/> " to go to the official Python website's Downloads section.
3. Through the magic of coding, the website will probably know what type of computer you are using, and the **DOWNLOAD** button will show you the correct version of Python to install! In our case, we want the latest version, which was Python 3.7.0 when I wrote the book. Don't worry if it tells you to download a newer version. You can also find the installer for your specific machine in the Files section.



Python Downloads (64-bit) - www.python.org

Files

Version	Operating System	Description	MD5 Sum	File Size	32-bit
Python 3.7.0 (64-bit)	macOS 10.9+	Python 3.7.0 (64-bit) for macOS 10.9+	4d90d30d44f103a0717030a0d211	33.7 MB	Yes
Python 3.6.5 (64-bit)	macOS 10.9+	Python 3.6.5 (64-bit) for macOS 10.9+	4d90d30d44f103a0717030a0d211	33.7 MB	Yes
Python 3.6.4 (64-bit)	macOS 10.9+	Python 3.6.4 (64-bit) for macOS 10.9+	4d90d30d44f103a0717030a0d211	33.7 MB	Yes
Python 3.6.3 (64-bit)	macOS 10.9+	Python 3.6.3 (64-bit) for macOS 10.9+	4d90d30d44f103a0717030a0d211	33.7 MB	Yes
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4. After clicking on the version, a download should start. Wait for it to finish before starting the installer.
5. When you start the installer, you should see a window like this one:



6. Click the **CONTINUE** button. You'll then be presented with some important information that you can choose to read or not.